

Peter Werner

[LinkedIn](#) | [Google Scholar](#) | [wernerpe.github.io](#) | wernerpe@mit.edu

EDUCATION

Massachusetts Institute of Technology (MIT) <i>Ph.D. in electrical engineering and computer science in Prof. Daniela Rus' group</i>	Cambridge, MA Sep. 2021 – ...
Eidgenössische Technische Hochschule Zürich (ETHZ) <i>Bachelor in Mechanical and Process Engineering, focus on mechatronics</i> <i>Masters in Robotics, Systems and Control, focus on learning and control.</i>	Zürich Sep. 2016 – May 2020 Sep. 2020 – May 2021
University of Pennsylvania (UPenn) <i>Exchange Semester at UPenn counting towards my MSc degree at ETHZ.</i>	Philadelphia Jan. 2020 – May 2020
Gymnasium and Military Service <i>Gymnasium Kirchenfeld Matura in Physics and applied Mathematics, one year military service</i>	Bern and Zürich region Aug. 2013 – May 2016

PROJECT AND WORK EXPERIENCE

Research Intern at Toyota Research Institute <i>Cuda & Convex Optimization</i>	May 2024 – Aug. 2024
Summer internship with the mobile manipulation team at the Toyota Research Institute where I developed the real-time motion planning pipeline published in [1].	
Grad. Research Assistant <i>MARL, Combinatorial & Convex Optimization</i> Prof. D. Rus	Sep. 2021 – present
Research on applied multi-agent reinforcement learning for multi-robot coordination, and collision-free motion planning.	
ANYmal learns Badminton <i>Python, C++, Reinforcement Learning</i> Prof. M. Hutter	Mar. 2021 – Sep. 2021
For my master's thesis I used deep reinforcement learning in Nvidia Isaac Gym to train ALMA C (ANYmal with arm) to play a simplified version of badminton. (Robotic Systems Laboratory, ETHZ)	
Vision-Based Sensing <i>Python, C++, Computer Vision</i> Prof. R. D'Andrea	Jan. 2019 – Sep. 2019
For my bachelor's thesis I developed and implemented vision-based proprioceptive sensing for an inflatable linear soft actuator at the Institute for Dynamical Systems and Control (ETHZ). → Paper, Video	
Research Assistant <i>Python, C++, ROS, Optimization, Machine Learning</i>	Sep. 2019 – May 2020
- Modeling of residual dynamics of VoliroX drone using Gaussian Process Regression and Locally Weighted Projection Regression at the Autonomous Systems Lab (ETHZ). My work included the implementation of an optimization based compensation scheme to reduce the effects of the residual dynamics. - Implementation of a model predictive contouring controller for autonomous racing on the F1TENTH platform at mLab, UPenn: Github	
Undergraduate Teaching Assistant	Sep. 2018 – May 2019
- Held weekly recitations for courses in Mechanics, Dynamics and Quantum Mechanics for roughly 30 students - Was selected as one of the best TAs for Quantum Mechanics and asked to hold an exam preparation course	

AWARDS & PRIZES

2x Outstanding D-MAVT Bachelor Award (1st year, overall)	ETH Zürich, Sep. 2017 & 2020
Received once for achieving one of the top 5 grade averages out of 543 students on the first year examinations and a second time for graduating with one of the top 5 GPAs out of 262 graduates.	
SGA Förderpreis	SGA, Nov. 2019
Award for the best Bachelor's Thesis in the field of Automatic Control issued by the Swiss Society for Automatic Control (Schweizerische Gesellschaft für Automatik, SGA)	
Excellence Scholarship & Opportunity Programme	ETH Zürich, Mar. 2020
Merit-based full-ride scholarship throughout the whole Master's program, awarded by the rector of ETH.	

WAFR 2022 Best Paper Award

WAFR, Jun. 2022

Award for the best paper accepted to the 15th international Workshop on the Algorithmic Foundations of Robotics (WAFR). Title: *Finding and Optimizing Certified, Collision-Free Regions in Configuration Space for Robot Manipulators*.

ICRA 2023 Best Workshop Paper Award

ICRA, Jun. 2023

Award for the best paper at the Multi-Robot Learning workshop at ICRA 2023. Title: *Dynamic Multi-Team Racing: Competitive Driving on 1/10-th Scale Vehicles via Learning in Simulation*

PUBLICATIONS

- [1] **Peter Werner**, Richard Cheng, Tom Stewart, Russ Tedrake, and Daniela Rus. “Superfast configuration-space convex set computation on GPUs for online motion planning”. In: *Robotics, Systems, and Science (RSS)*. 2025.
- [2] Alexandre Amice, **Peter Werner**, and Russ Tedrake. “Certifying bimanual rrt motion plans in a second”. In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2024, pp. 9293–9299.
- [3] **Peter Werner**, Alexandre Amice, Tobia Marcucci, Daniela Rus, and Russ Tedrake. “Approximating robot configuration spaces with few convex sets using clique covers of visibility graphs”. In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2024, pp. 10359–10365.
- [4] **Peter Werner**, Thomas Cohn, Rebecca H Jiang, Tim Seyde, Max Simchowitz, Russ Tedrake, and Daniela Rus. “Faster Algorithms for Growing Collision-Free Convex Polytopes in Robot Configuration Space”. In: *International Symposium of Robotics Research (ISRR)*. 2024.
- [5] Hongkai Dai*, Alexandre Amice*, **Peter Werner**, Annan Zhang, and Russ Tedrake. “Certified polyhedral decompositions of collision-free configuration space”. In: *International Journal of Robotics Research* (2023).
- [6] **Peter Werner***, Tim Seyde*, Paul Drews, Thomas Matrai Balch, Igor Gilitschenski, Wilko Schwarting, Guy Rosman, Sertac Karaman, and Daniela Rus. “Dynamic Multi-Team Racing: Competitive Driving on 1/10-th Scale Vehicles via Learning in Simulation”. In: *7th Annual Conference on Robot Learning*. 2023.
- [7] Alexandre Amice*, Hongkai Dai*, **Peter Werner**, Annan Zhang, and Russ Tedrake. “Finding and optimizing certified, collision-free regions in configuration space for robot manipulators”. In: *International Workshop on the Algorithmic Foundations of Robotics*. Springer. 2022, pp. 328–348.
- [8] Tim Seyde, **Peter Werner**, Wilko Schwarting, Igor Gilitschenski, Martin Riedmiller, Daniela Rus, and Markus Wulfmeier. “Solving Continuous Control via Q-learning”. In: *The Eleventh International Conference on Learning Representations*. 2022.
- [9] **Peter Werner**, Matthias Hofer, Carmelo Sferrazza, and Raffaello D’Andrea. “Vision-based proprioceptive sensing: Tip position estimation for a soft inflatable bellow actuator”. In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE. 2020, pp. 8889–8896.

TECHNICAL SKILLS

Languages: Python, C/C++, Matlab, some CUDA C++

Frameworks and Libraries: ROS, pandas, NumPy, Matplotlib, OpenCV, pyTorch, keras, CASADI, ForcesPRO, gym, isaacgym, Drake, CVXPY, Gurobi, Mosek

Developer Tools: Git, Docker, Visual Studio, PyCharm, CMake, Bazel

Engineering Tools: Linux, Solidworks, Siemens NX, Latex, MS Office

ADDITIONAL

Languages: English (native), German (proficient), Swiss German (native), French (working proficiency)

Interests: Running, hiking, badminton, skiing, climbing, and comics